Sequential Logic – Part 1
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Review

- We have discussed combinational logic.
- We will now begin discussion of various sequential logic circuits.

Why are sequential logic circuits important ?

- Various gadgets that we use today (personal computers, mobile phones etc.) not only perform computation (eg. adders we have discussed in combinational logic) but also store information.
- The second part, namely storage, is what makes study of sequential logic circuits important.

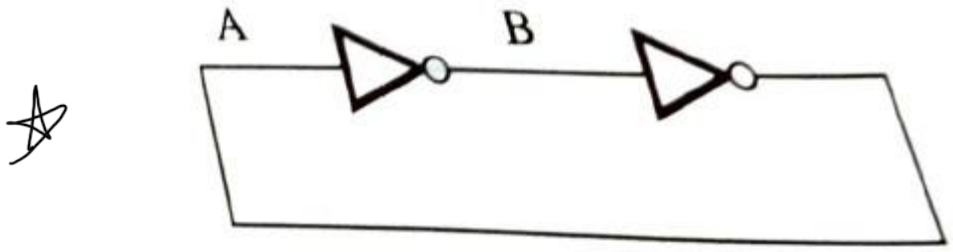
What is a sequential logic circuit ?

- We observed that a combinational logic circuit is one whose outputs at any time depend on only the current combination of inputs.
- In contrast, in a sequential logic circuit, the values of outputs depend not only on the present values of the inputs but also on the past behaviour of the circuit. We will discuss more about behaviour shortly.
- Sequential circuits include storage elements that store the values of the logic signals. The contents of the storage elements represent the state of the circuit.

More on sequential logic circuits

- When the circuit's inputs change (values), the new input values may leave the circuit in the same state or cause it to move to another state.
- Over time, the circuit goes through a sequence of states due to changes in inputs.
- Circuits that have this behaviour are called sequential circuits.

Very simple storage element



How does it work ?

- If we assume A = 0, then B = 1 and vice-versa.
- So the circuit will maintain these values indefinitely.
- We can say that the circuit is in the state defined by these values.
- What is the problem ?

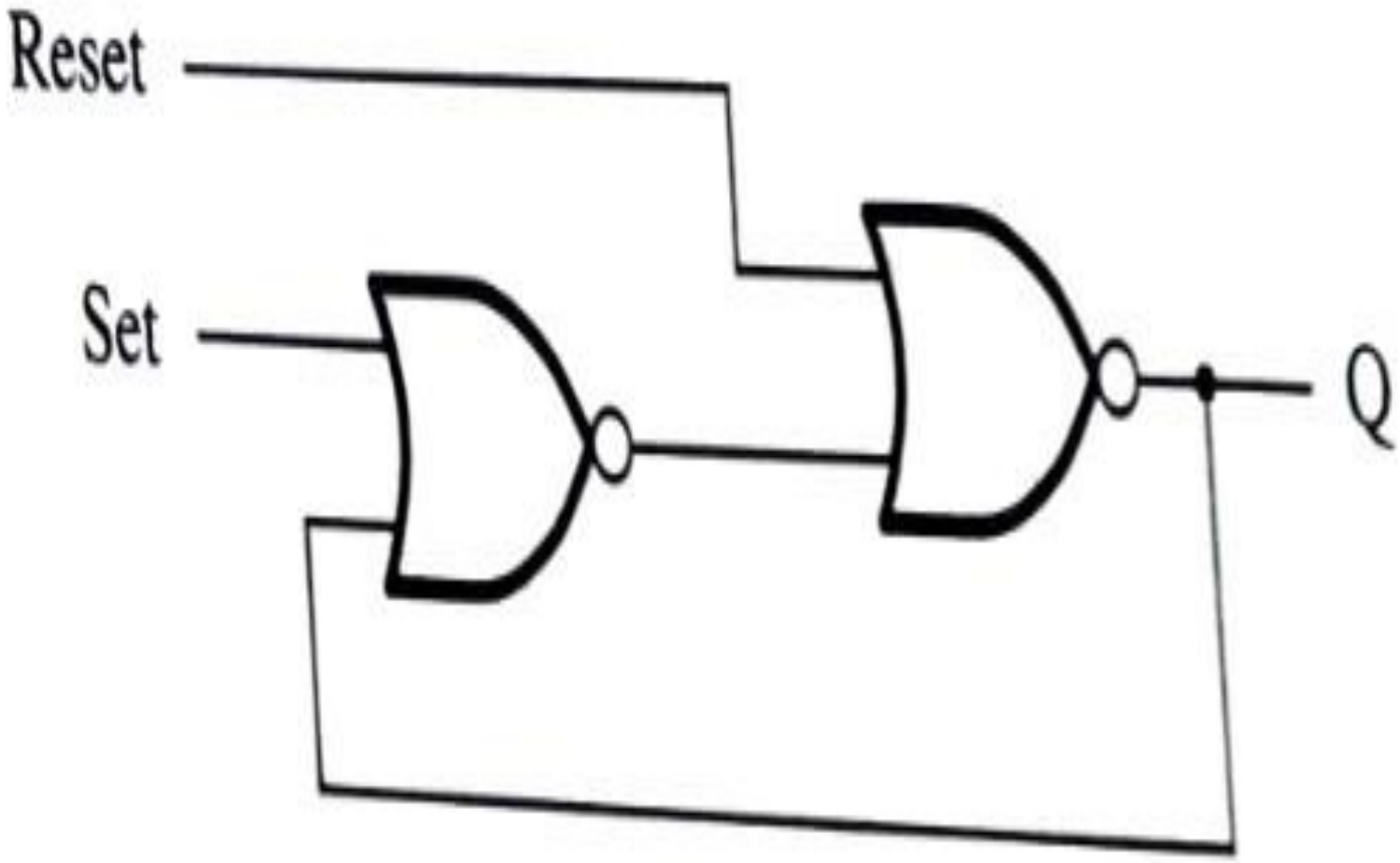
What is the problem ?

- There is no means to change the state.
- So we need something a little more sophisticated.
- This leads to what are called latches.
- We will start with a basic latch and then look at modifications.

Latch

- The basic latch is one that supports set and reset. It is called as an SR latch.
- The SR latch can be constructed with a pair of NOR gates or with a pair of NAND gates.
- We will show an arrangement with a pair of NOR gates next.

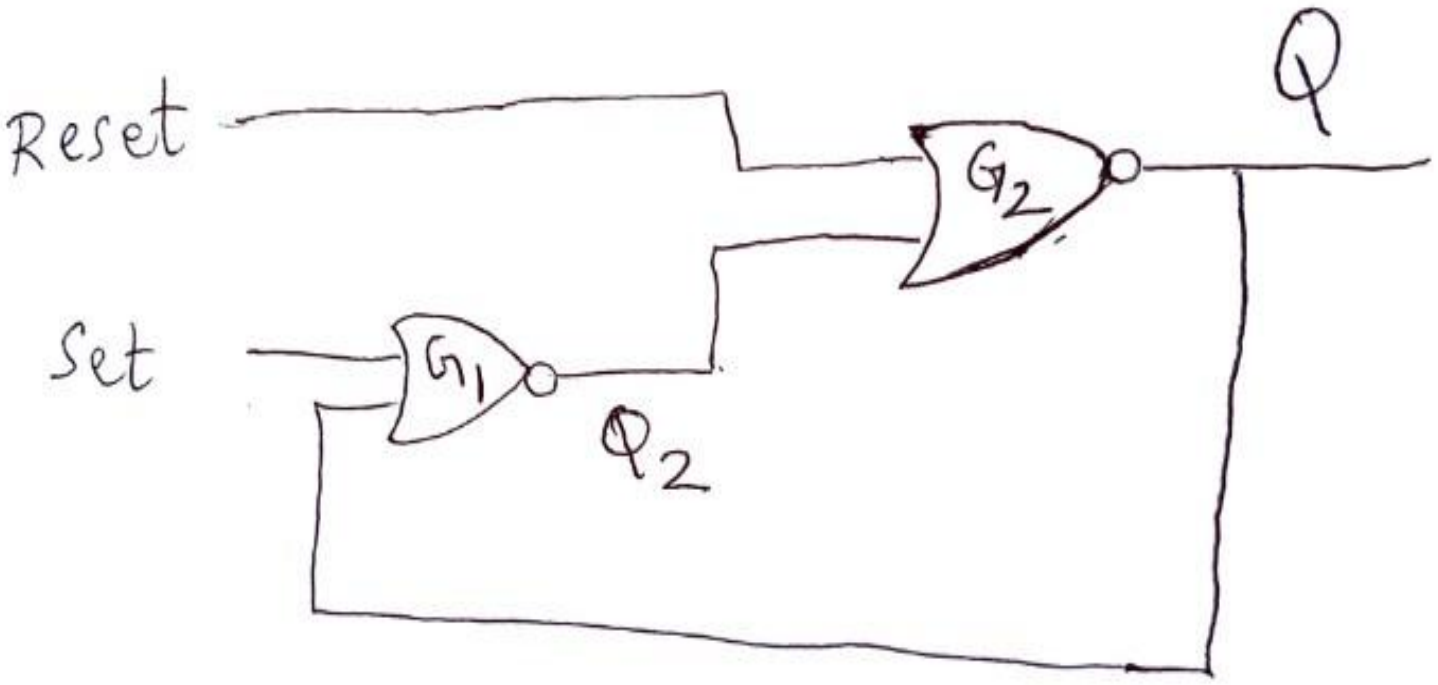
SR latch with NOR gates



What does this latch do ?

- When the Set input is high, the latch is set. That is, Q will become 1 when Set = 1.
- Assuming Reset is low, even if we change Set to 0 from 1, Q will still remain 1.
- So it stores the values of 1.
- When we change Reset to high, with Set remaining at 0, the latch will be reset (meaning Q will become 0). Thereafter, even if Reset = 0 (with Set remaining 0), Q will remain 0.

The same circuit with extra labels



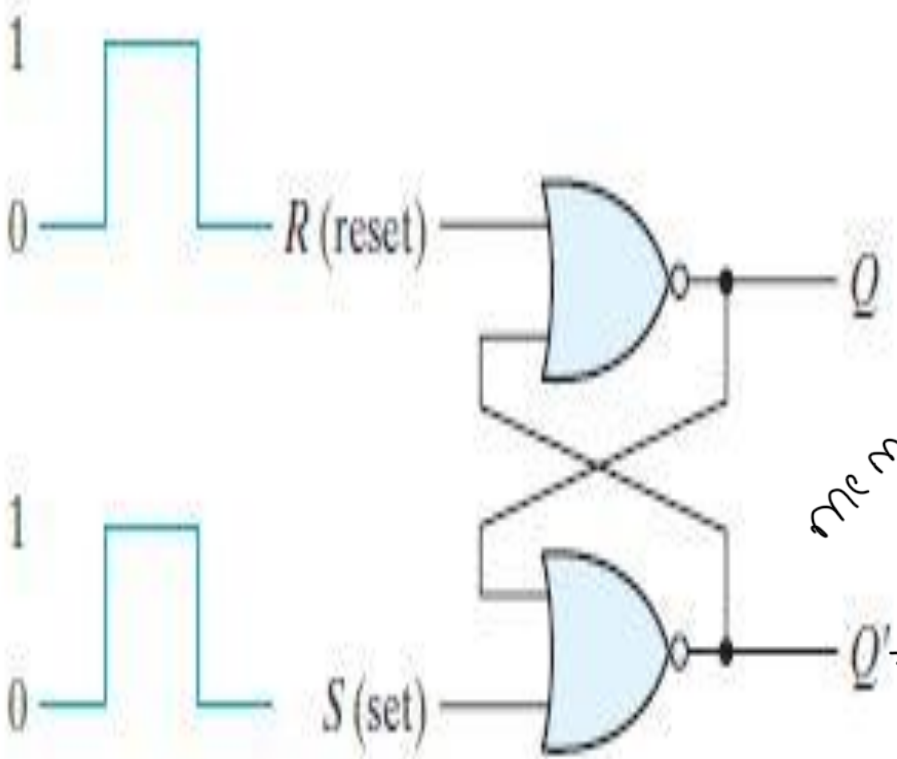
Analyzing the circuit more ..

- How does the feedback work ?
- Suppose Q is 0 initially when Set becomes 1. We note that Q2 will be 0. If Reset is 0, then we note that gate G2 will set Q to 1.
- Suppose Q had been 1 initially when Set becomes 1. Then Q2 will still be 0. If Set goes to 0 with Q = 1, once again Q2 will be 0.
- If Q2 is 0 and Reset becomes 1, then Q will become 0.

By the way ..

- We assume power is delivered to the IC that contains this latch circuit.
- That is V_{cc} and Gnd (ground) pins are connected appropriately.
- We now show the usual way the SR latch is depicted in texts. It shows the NOR gates connected in cross-coupled style.

SR latch with NOR gates



(a) Logic diagram

memory

S	R	Q	Q'	
1	0	1	0	
0	0	1	0	(after $S = 1, R = 0$)
0	1	0	1	
0	0	0	1	(after $S = 0, R = 1$)
1	1	0	0	(forbidden)

(b) Function table

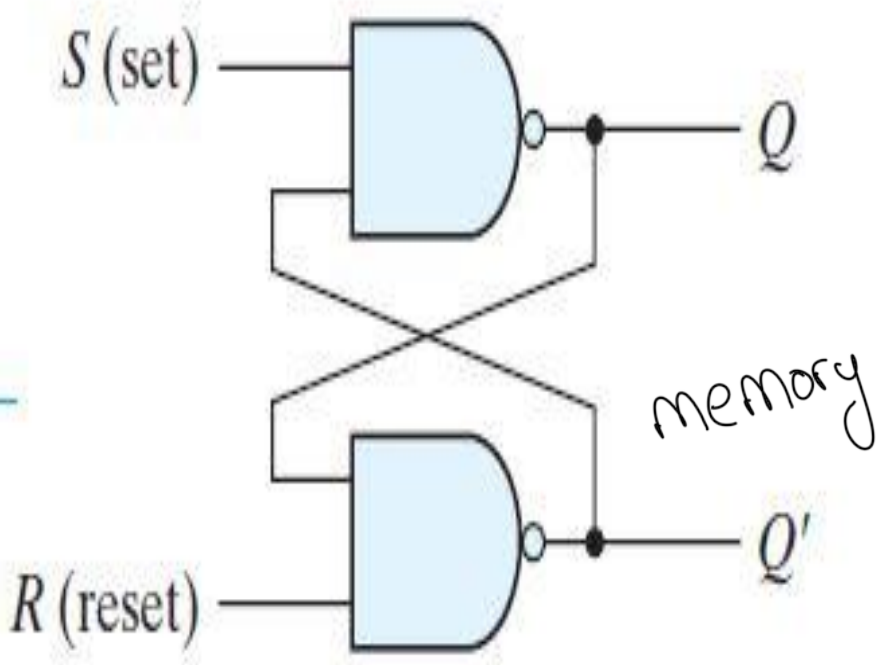
Explanation

- The SR latch operation is shown by the function table. Initially, both the inputs are 0. When S is set to 1, the output Q = 1 and Q’ = 0. We need to change S to 0 before applying R (or set R to 1) to avoid an undefined next state as explained below.
- When both S and R are 1 at the same time, both the outputs become 0. This violates the basic rule of latch operation that requires the latch outputs to be complements.

More on the SR latch

- Further, suppose both S and R are 1 at the same time. If we now switch the inputs to 0 simultaneously, the device will enter an unpredictable state (that depends on the order in which the transitions happen).

What if we use a pair of NAND gates ?



(a) Logic diagram

S	R	Q	Q'	
1	0	0	1	
1	1	0	1	(after $S = 1, R = 0$)
0	1	1	0	
1	1	1	0	(after $S = 0, R = 1$)
0	0	1	1	(forbidden)

(b) Function table

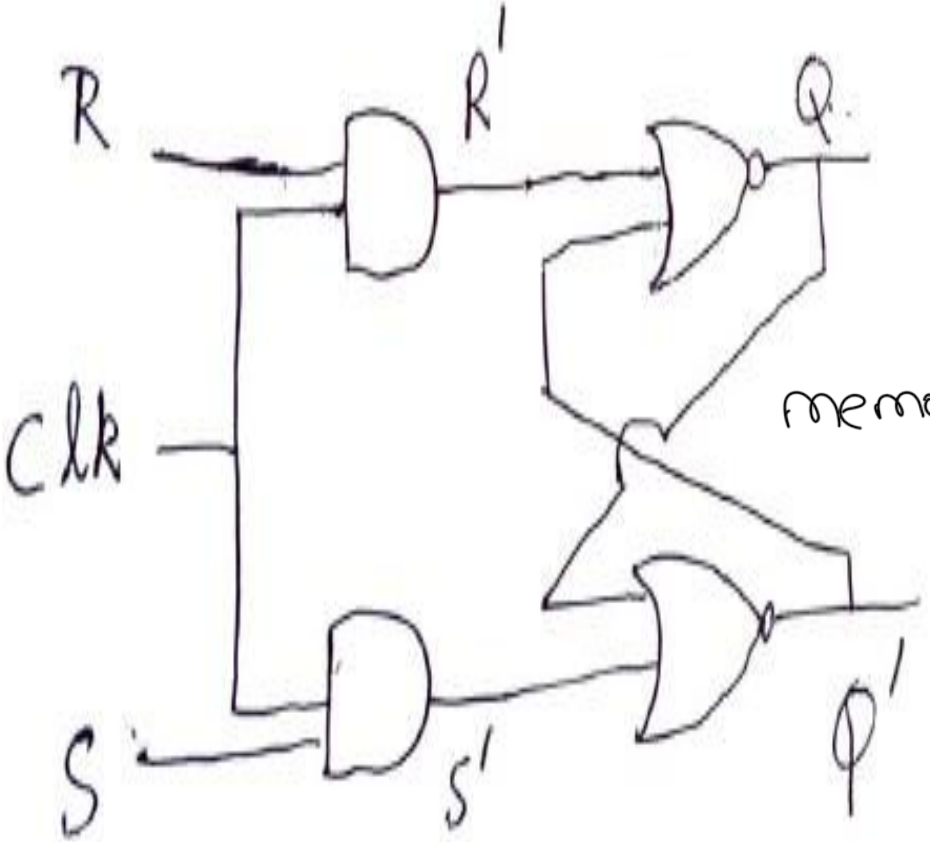
SR latch with a pair of NAND gates

- Instead of applying 1 to S, apply 0 to S (with R =1) to get Q to 1 (this puts latch in set state).
- When S goes to 1 now, the latch will remain set.
- To reset the latch, make R = 0 (with S remaining at 1).
- Here, S=R = 0 is forbidden.

Is there a way to decide when the state of the SR latch should change ?

- Yes, this is where the clock (Clk) signal comes in.
- It may also be referred to as an enable signal.
- The entire arrangement is sometimes called as a gated SR latch.

SR latch with enable (Clk) signal



Clk	S	R	Next state of Q
0	x	x	No change
1	0	0	No change
1	0	1	0
1	1	0	1
1	1	1	Indeterminate

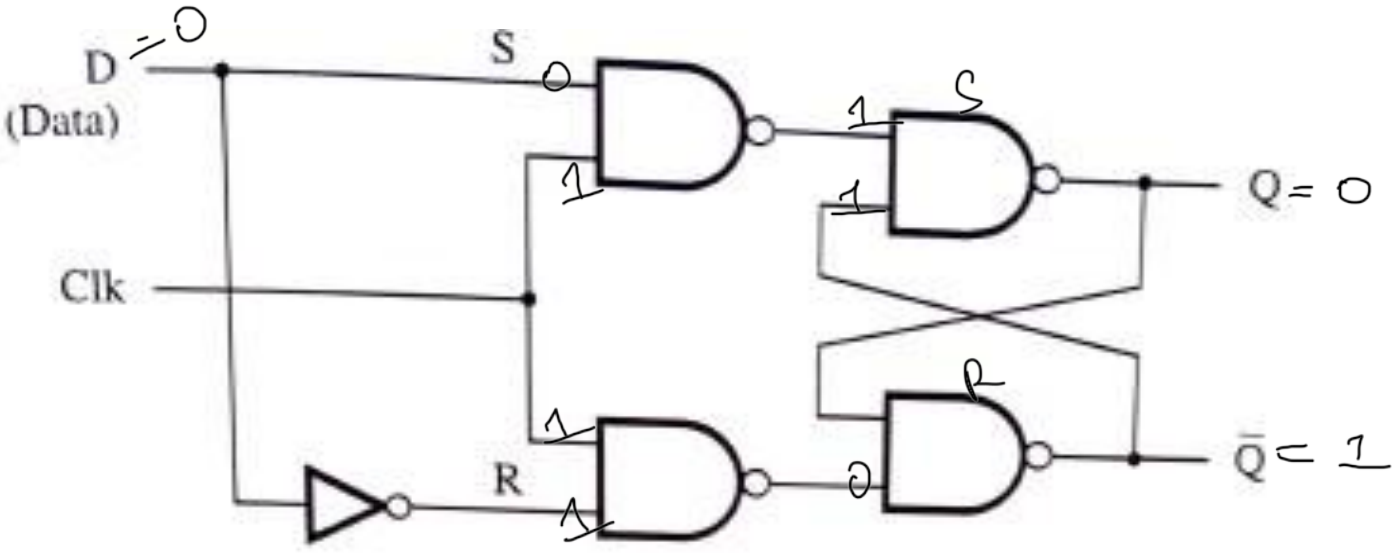
Explanation

- The clock input determines when the state of the latch can be changed.
- That is, the outputs of the (left) AND gates stay at logic 0 as long as the Clk signal remains at 0 (irrespective of what R and S are).
- When Clk = 1, inputs S or R affect the latch.
- When all the three inputs (Clk, R and S) are 1, we have an indeterminate condition since one of the inputs to both the right side gates becomes 1.

How do we handle the indeterminate state ?

- One solution is to use what is called as a D (data) latch. In this case, we do not have explicitly three inputs (as before).
- Instead, we have only 2 inputs, namely the D input and the Clk input.
- The structure of the D latch with an enable (Clk) input is shown next.

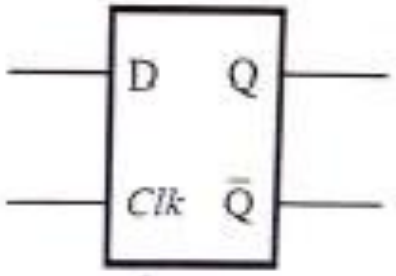
D latch with Clk input



(a) Circuit

Clk	D	Q(t + 1)
0	x	Q(t) → memory
1	0	0
1	1	1

(b) Truth table



(c) Graphical symbol

Explanation

- Note that the D (data) input goes to S (set) input while its complement goes to R.
- When Clk=0, the left side NAND gates will output 1 (regardless of the D input) so the overall circuit outputs (Q and Qbar) will not change. Verify this by assigning Q = 0 and see what happens to Q' (Qbar); then set Q = 1 and check Qbar. No change is also indicated by Q(t + 1) being equal to Q(t).

More on D latch with Clk (gated D latch)

- When the Clk input becomes 1, the D input value determines Q.
- In particular, when Clk= 1 and D is 1, the latch is set (i.e., Q = 1). Just look at the right side NAND gate that gets a 0 input and infer the Q output (do not get confused by the input to the other right side NAND gate)
- When D is 0, with Clk = 1, the latch gets reset (i.e., Q = 0).

A note on timing diagrams

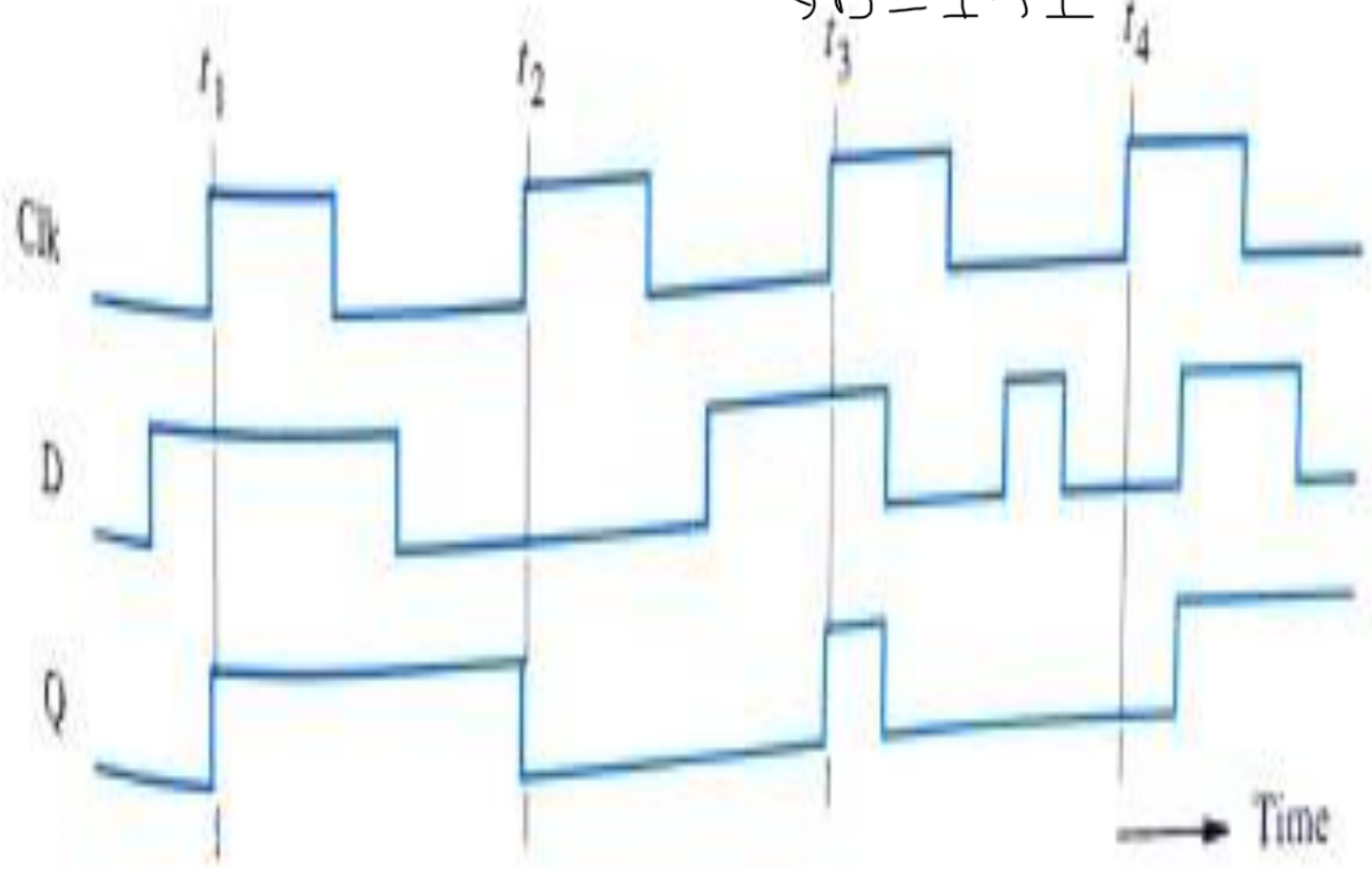
- Besides getting the truth table (the next state table here), we can also show graphically how the signals are.
- This is called as a timing diagram. We will show the timing diagram of the gated D latch next.

Timing diagram of gated D latch

clock → 0 → memory

clock → 1 → D = 0 → 0

→ D = 1 → 1



On the timing diagram

- What we have shown is the manner in which the output (Q) changes in response to changes in the D (data) input as well as the clock (Clk) input.
- Notice when Q changes corresponding to changes in D and for how long Q remains that way.
- Since the output of the gated D latch is controlled by the level of the clock input, the latch is said to be level sensitive.

What happens if D changes at the same time as the Clk ?

- How do we know what Q will be ?
- This is an important thing to look at: will the change at D be reflected at Q ?
- We will discuss this later with reference to some quantities. In particular, we will introduce terms called setup time and hold time.

Textbook and Reference

- M. Mano and M. Ciletti, Digital Design, Pearson Education, 2018
- S. Brown and Z. Vranesic, Fundamentals of digital logic with VHDL design, McGraw Hill, 2017.
- C. Roth and L. Kinney, Fundamentals of Logic Design, Cengage Learning, 2014.